

Topological Analysis of Plant Structure Using Persistent Homology for Genotype Classification

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Outline

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Introduction

- **Plant architecture** determines how genetic traits translate into growth, development, and stress resilience.
- Accurate **morphological characterization** supports both research and crop improvement.
- **Morphological and skeleton-based features** offer valuable insights, especially with **deep learning** integration.
- However, these approaches rely on **predefined shape assumptions** and fixed spatial scales → sensitive to **noise, occlusion, and deformation**.
- These descriptors often overlook **the global structural integrity** of plant architecture or repeated motifs that reflect **complex growth patterns**.

Existing Morphological and Skeleton-Based Features

- Using **RGB Images**, various Morphological and Skeletal Features were extracted to get insights into plant architecture. Features such as :
- **Convex Hull Area** which refers to the area of the smallest convex polygon that fully encloses the plant or object.
- **Solidity** measures how completely the shape fills its convex hull. It is calculated as:

$$\text{Solidity} = \frac{\text{Object Area}}{\text{Convex Hull Area}}$$

- **Euclidean length** is the straight-line distance between two endpoints of a skeleton segment.
- **Eccentricity** quantifies how elongated an object is based on the best-fitting ellipse.

$$\text{Eccentricity} = \sqrt{1 - \frac{b^2}{a^2}}$$



Topological Data Analysis (TDA)

- **TDA** provides a mathematical framework to study the **shape and structure** of complex data.
- It focuses on **how data points connect and form patterns** across multiple scales, without relying on coordinate systems.
- Instead of analyzing isolated points, TDA identifies **global structural features** — clusters, loops, and voids — that persist across spatial resolutions.
- **Persistent Homology (PH)** captures:
 - *Connected components (0D)*
 - *Loops or holes (1D)*
 - *Voids (2D)*
- These features **emerge, disappear, and vary** as the **spatial resolution or filtration parameters** of images change.

Motivation

- Traditional skeleton and morphological descriptors **miss global structural patterns**.
- There is a lack of a **scale-invariant, noise-resilient** descriptor for RGB-based morphology.
- Persistent Homology(PH) captures **connections and shape evolution** across scales.
- It encodes **global complexity** (loops, voids, components) in an **interpretable vector form**.
- Unlike geometric metrics, it is **robust** to noise, deformation, and resolution — ideal for high-throughput phenotyping.
- Combining **PH-derived topological vectors** with ML models can enhance **genotype classification** and **phenotypic analysis**.

Role of Persistent Homology in Our Framework

- **Persistent Homology (PH)** can be applied to both **static** and **timeseries** plant imaging data, though its utility differs significantly across these scenarios.
- In **time-series data**, PH tracks *topological feature evolution* — emergence/disappearance of loops and components → reveals **growth dynamics** and **stress responses**.
- From a **single image**, PH quantifies *canopy complexity* and *internal voids*, offering static structural descriptors.
- Therefore, while PH is useful in both contexts, its full potential is best realized with **multi-day time-series** data that reflect the **dynamics of plant architecture**.

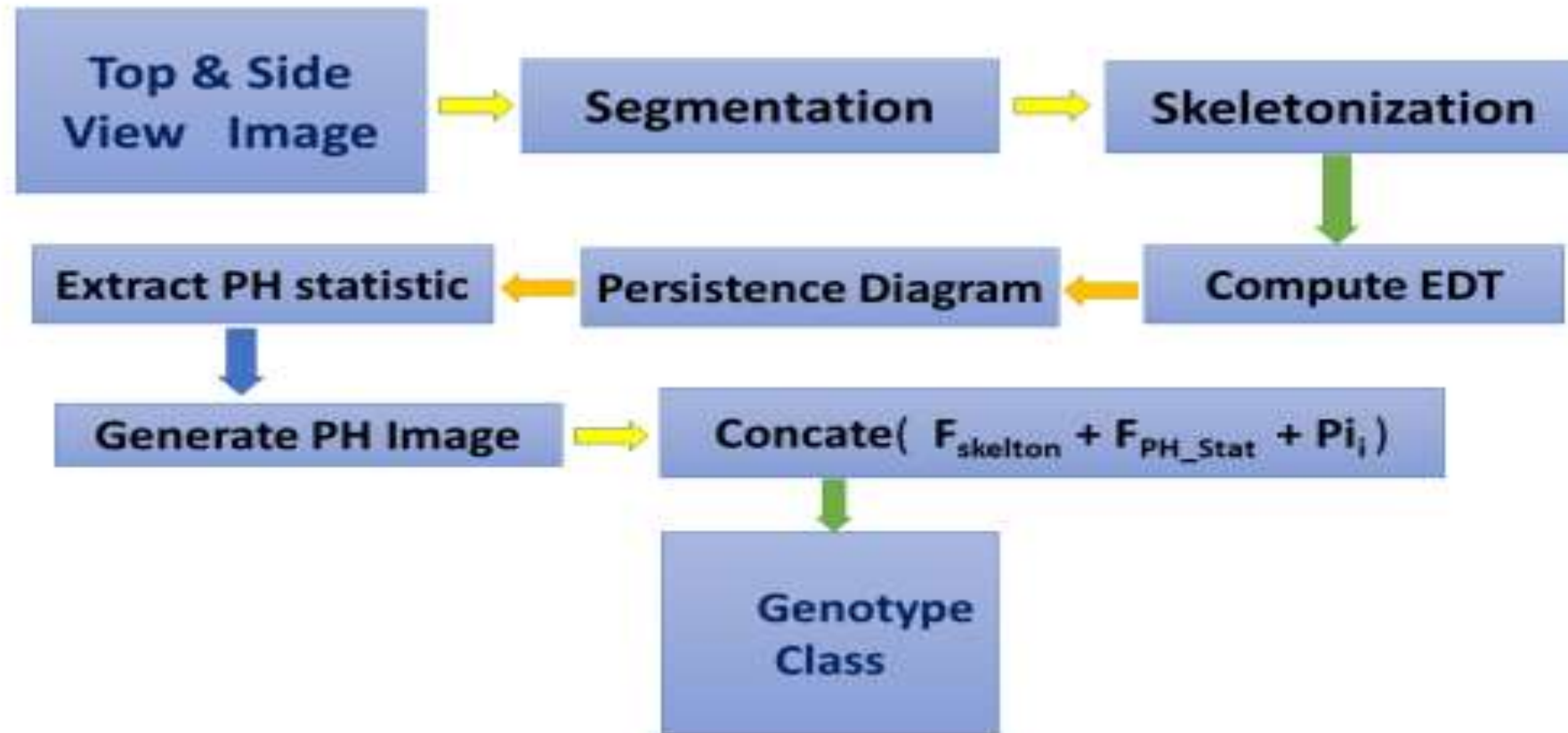
Classification Models Employed

- **Random Forest**: Random Forest is an ensemble learning algorithm that builds multiple decision trees during training and combines their results to make a final prediction.
- **Multilayer Perceptron (MLP)**: MLP is a type of neural network made up of multiple layers of connected neurons. It learns to map input features to the correct output class by adjusting the weights of connections through a process called backpropagation.
- **Support Vector Machine (SVM)**: Support Vector Machine is a supervised learning method used for classification tasks. It works by finding the best possible boundary—called a hyperplane—that separates data points from different classes.

OBJECTIVES

- **To develop** a framework for **quantitative characterization of plant morphology** using RGB images.
- **To overcome** limitations of traditional geometric and skeleton-based features by integrating **topological data analysis (TDA)**.
- **To extract** robust and scale-invariant **topological descriptors** using **Persistent Homology (PH)**.
- **To combine** PH-based features with **morphological descriptors** for improved **genotype classification**.
- **To evaluate** and compare the performance of different **machine learning models** (SVM, Random Forest, MLP) using the proposed feature set.

Proposed Pipeline



Experimental Dataset

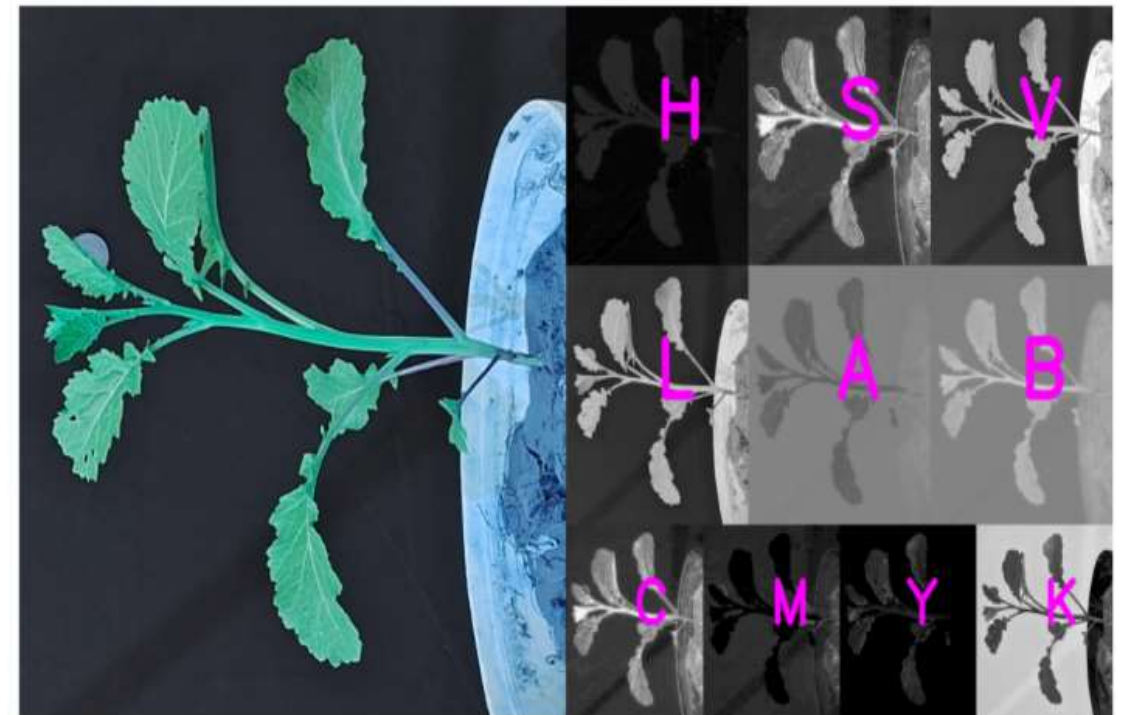
- Dataset of **10 Brassica juncea (Indian mustard) genotypes** was used
(*RE-2-2, RE-8, RE-13, RE-14, RE-35-4, CN-101849, CN-105306, CN-105308, CN-101846, CN-112920*)
- Plants imaged from **both side and top views** to capture diverse morphological traits.
- Images acquired under **uniform conditions** with a maintained resolution (*e.g., 1024×768 px*).
- Dataset is **balanced across genotypes**, supporting reliable **feature extraction** and **classification**.



Methodology

- RGB plant images were preprocessed and segmented using the **PlantCV** library.
- Converted to multiple **colour spaces** to extract diverse visual features:
 - **HSV** – encodes hue, saturation, and brightness.
 - **CIELAB** – aligns with human vision; separates lightness (L^*) from colour (a^* , b^*).
 - **CMYK** – represents colours through subtractive ink model (cyan, magenta, yellow, black).

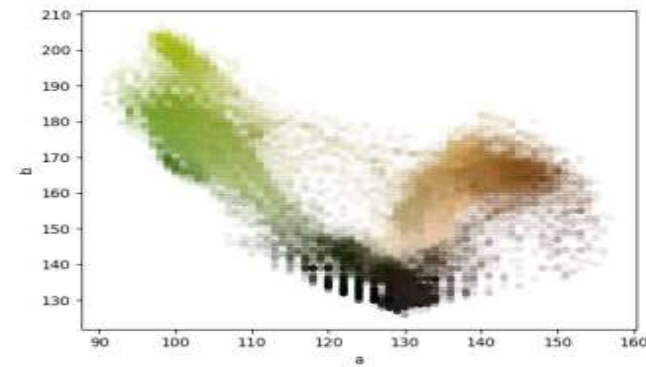
These conversions enhance **structural and colour-based segmentation** for downstream feature extraction.



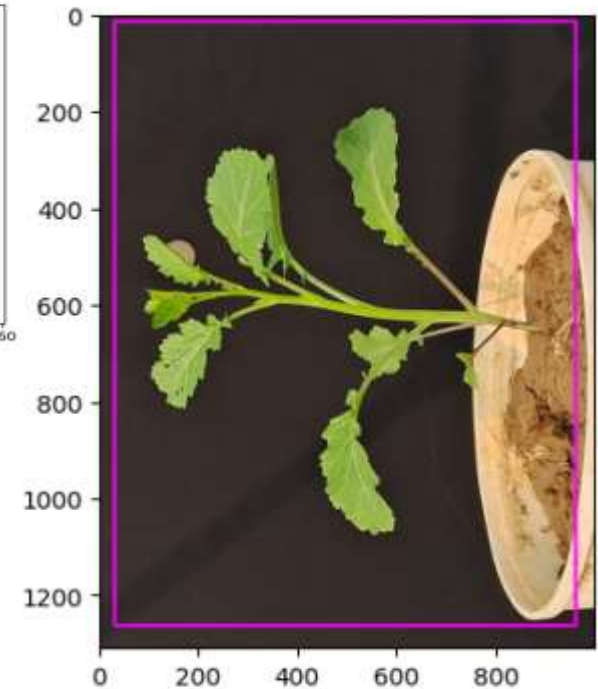
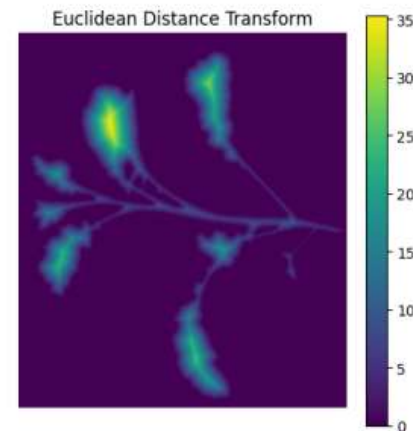
Segmentation and ROI Extraction

- **Binary Thresholding:** Isolated plant structure from background using the **b*** channel of the LAB colour space.
- **Region of Interest (ROI):** Defined a rectangular area to focus analysis and reduce background noise.
- **Euclidean Distance Transform (EDT):** Applied on the plant mask M_i to create a grayscale distance map D_i for topological feature extraction.

$$D_i(x, y) = \min_{(x', y') \in \text{background}} \sqrt{(x - x')^2 + (y - y')^2}.$$



Scatter plot



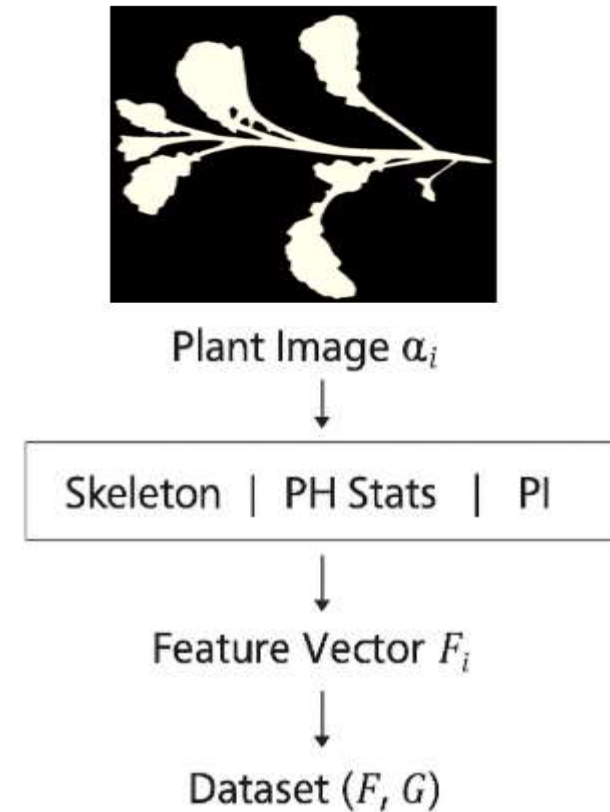
ROI

Topological Filtration and Persistence Diagram Construction

- A **topological filtration** is built from the Euclidean Distance Transform (D_i) using a sequence of **superlevel sets** ($F_t = \{(x, y) \mid D_i(x, y) \geq t\}$), gradually revealing the plant's shape.
- The image is treated as a **2D cubical complex**, where each pixel represents a topological cell contributing to nested structural layers.
- Using this filtration, a **Persistence Diagram (PD_i)** is computed — capturing the **birth and death** of topological features (connected components, loops) as t varies.
- Each point (b_j, d_j) in the diagram represents a feature's lifespan; **large persistence** ($d_j - b_j$) indicates significant structure, while **short-lived ones** often represent noise.
- From these persistence diagrams, **a range of quantitative topological features** (e.g., counts, persistence measures, and loop statistics) were extracted for downstream analysis.

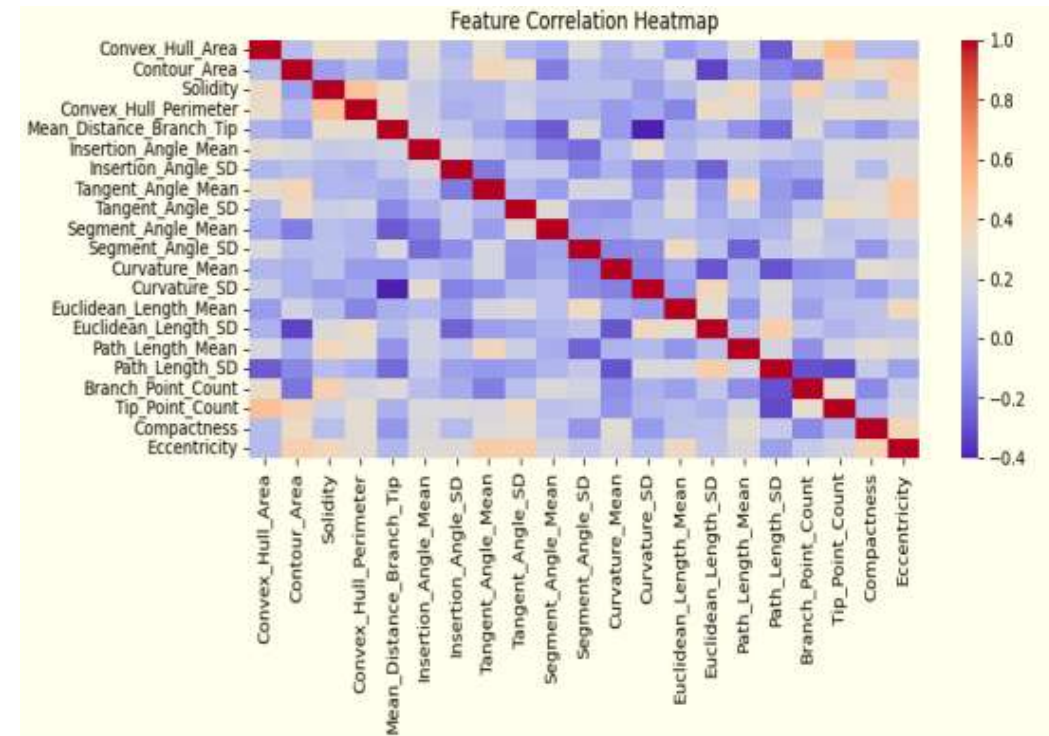
Feature Integration and Dataset Formation

- Each plant image (α_i) produces multiple feature components $\rightarrow$ combined into a single vector F_i .
- The vector F_i integrates three types of features:
 - Skeleton-based structural descriptors (F_{skel})
 - Persistent homology summary statistics (F_{PHstat})
 - persistence image (PI_i , 900-D Flattened Vector)
- All F_i vectors form dataset $F \in \mathbb{R}^{n \times 931}$
- Each vector linked with genotype label g_i , forming supervised set (F, G) for ML models



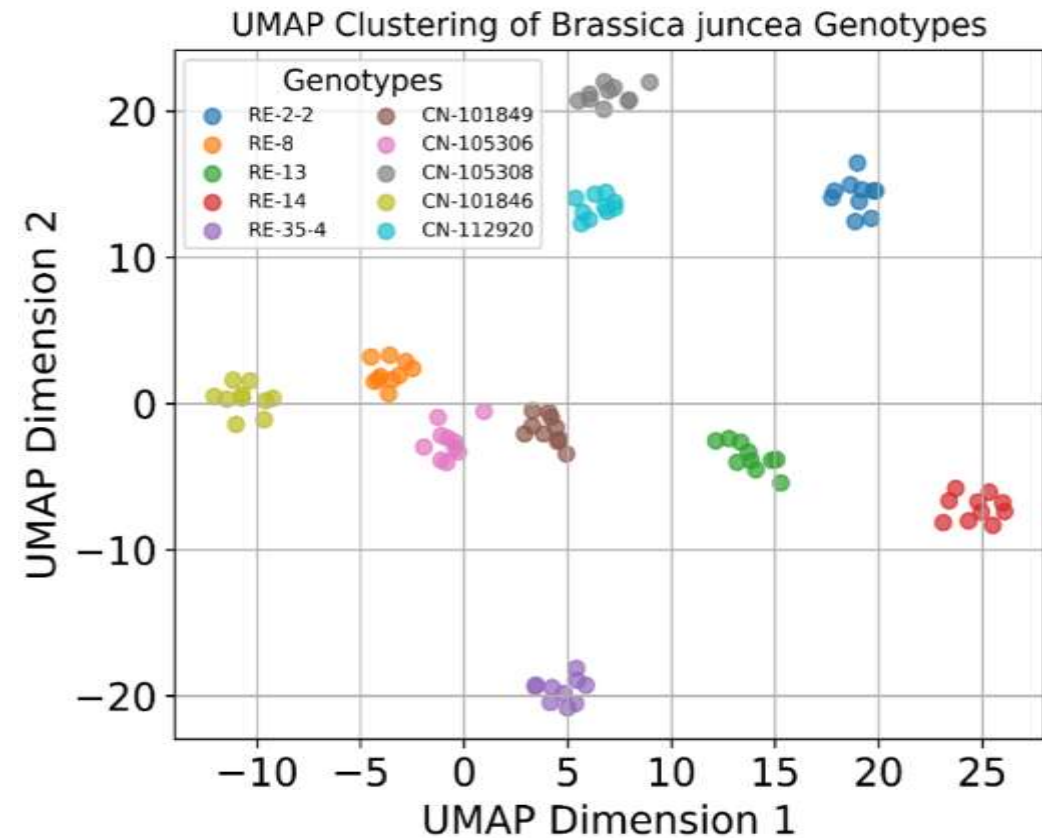
Experimental Results

- **Random Forest:** The Feature Importance is based on **mean decrease in Gini impurity** or **mean decrease in accuracy** when features are permuted.
- **MLP (Neural Network):** Its Feature Importance is derived from **connection weight analysis** or **permutation importance**.
- **SVM:** For SVMs, importance is not straightforward to compute, especially with nonlinear kernels.



Experimental Result

- High-dimensional feature vectors were projected into 2D using **UMAP** for visualization.
- Distinct clusters represent **10 Brassica juncea genotypes**, showing that extracted features capture **genotype-specific structures**.
- Both **skeleton-based** and **topological (PH)** features enhance inter-genotype separation.
- Tight clusters → **intra-genotype consistency**
- Wide gaps → **distinct phenotypic traits**



Conclusion

- Integration of **Persistent Homology (PH)** with **skeleton-based** features significantly improves genotype classification.
- **Skeleton features** capture fine structural traits, but miss global topology.
- **PH features** capture large-scale connectivity and shape complexity.
- **Combined features** provide complementary views → leading to higher **accuracy and F1-scores** across all models, especially **Random Forest and SVM**.

Classifier	Feature Set	Accuracy (%)	F1-score (%)
SVM	Skeleton only	87.4	86.6
SVM	Skeleton + PH	93.7	93.1
Random Forest	Skeleton only	88.5	87.9
Random Forest	Skeleton + PH	94.2	93.6
MLP	Skeleton + PH	92.5	91.1

Conclusion and Future Work

- While the proposed framework demonstrates strong classification performance and structural interoperability, several limitations remain.
- The dataset used in this study is limited in scale, comprising a fixed set of genotypes captured under controlled imaging conditions, which may restrict its applicability to more variable field environments.
- Future work will aim to broaden the dataset by incorporating a wider range of genotypes and environmental conditions, along with temporal image sequences to enable the analysis of dynamic growth patterns and developmental trajectories.



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THANK YOU